

Minor_1 2025: Software Engineering

Time: 2 hours; Marks: 50

Instructions:

1. Read all questions very carefully.
2. Answers must be precise. Verbosity will lead to reduction in scores.
3. All questions are compulsory.
4. You are encouraged to make reasonable assumptions as part of your answer.
5. All assumptions must be clearly stated as part of the answer. Please remember, these assumptions make your answer unique.
6. You are encouraged to use diagrams, tables and examples to substantiate your answer.
7. Neatness is a virtue.

Best Wishes!

Imagine that you have to design a 'novel' cloud-based multi-device-synchronized multimodal file-hosting service.

Answer the following questions in sync with the provided essence cards and required competence (minimum level 2) of all stakeholders for the above scenario.

a. What will be the functional and the non-functional requirements of your system? Compare and contrast with existing services.	[5*3 = 15]
b. Draw Class, Use-case, and Sequence UML diagrams to describe your system. [Please make sure that your diagrams are self-descriptive.]	[5*3 = 15]
c. What SDLC technique(s) would you follow? Justify.	[5]
d. Which architecture pattern(s) will you be using? Justify.	[5]
e. Given your chosen SDLC and architecture pattern, how will you ensure that the requirements in the provided essence cards will be met?	[10]

[Alpha cards on next page]

Stakeholders

Recognized

- ☐ Stakeholder groups identified
- ☐ Key stakeholder groups represented
- ☐ Responsibilities defined

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Stakeholders

Represented

- ☐ Responsibilities agreed
- ☐ Representatives authorized
- ☐ Collaboration approach agreed
- ☐ Way of working supported & respected

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Work

Initiated

- ☐ Required result clear
- ☐ Constraints clear
- ☐ Funding stakeholders known
- ☐ Initiator identified
- ☐ Accepting stakeholders known
- ☐ Source of funding clear
- ☐ Priority clear

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Software System

Architecture Selected

- ☐ Architecture selection criteria agreed
- ☐ HW platforms identified
- ☐ Technologies selected
- ☐ System boundary known
- ☐ Decisions on system organization made
- ☐ Buy, build, reuse decisions made
- ☐ Key technical risks agreed to

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Software System

Demonstrable

- ☐ Key architectural characteristics demonstrated
- ☐ System exercised & performance measured
- ☐ Critical HW configurations demonstrated
- ☐ Critical interfaces demonstrated
- ☐ Integration with environment demonstrated
- ☐ Architecture accepted as fit-for-purpose

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